Metascore: User Interface Design for Generative Film Scoring

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Abstract

This paper outlines the development of the Metascore application which is the result of investigations into visual interfaces for the control and coordination of parameters for generative film music scores. Metascore is part of a project under development at the Australasian CRC for Interaction Design (ACID), the aim of which is to create a real-time automatic film scoring system. Metascore allows the user to control generative music algorithms using high-level descriptions of compositional intent and in synchronisation with video. It provides control of constant parameters within sections that demarcate film cues, and control of continually varying parameters through free-form break point envelopes. Parameter values in Metascore can themselves be expressions or functions, enabling extensive flexibility for adaptive control of the algorithmic score. This paper will describe the features which we have found to be most useful in the composition of generative film music the design of the Metascore interface which exposes these features to the user.

Introduction

Film music provides both structural and emotive support to the narrative of a film. As a consequence, film music is constrained by the pace and structure of the film's narrative. Metascore is a film scoring and composition environment designed to make the setting and adjustment of video synchronised musical parameters quick and easy to manage. In combination with generative processes, Metascore is designed to make the composition process flexible and efficient, whilst providing the user with sufficient control over the development of the music to meet the expressive demands of the film narrative. Metascore's controls allow a score to be defined using descriptions of the video's structure, such as the timing of film cues and scene changes, and abstracted music descriptions, such as the harmonic tension or rhythmic density. These descriptions are then processed by Metascore to generate the film score.

When developing a film scoring tool, important considerations include the emotional content of the music and the time-based issues around synchronisation of music to video and visual events. This paper is primarily concerned with issues surrounding synchronicity and the design of a user interface to accommodate them; the compositional algorithms controlled by Metascore can vary from style to style and is discussed in related papers.

Design Considerations

When composing for film the integration of music and video needs be addressed. A general pattern in the film industry is that music must be adapted and shaped to fit the film. In the design of Metascore and its interface, we have considered two ways to integrate music with film.

Firstly, it is common practice for the score to emphasize and articulate the actions in the film, so that musical events coincide with visual events or actions. This may involve placing strong beats in the music to land on hit points, matching sectional changes in the music with scene changes, using crescendos to build up to a plot event, or other musical articulation of visual events. In order to accommodate these

cases, the tempo and time signatures used in the score must be carefully managed to ensure that the music is synchronised with the film. Given that film cues can fall at any interval, synchronisation attempts can lead to rather irregular musical structures. It can be a challenge for composers to comfortably fit the music around them to maintain the pace of the music.

A second scoring consideration is matching the emotional content of the music with the emotions in the film. The mood of the music must follow the film in an appropriate manner, usually done to achieve empathy or contrast with the intent to create a heightened collective experience. Adding to the challenge, the mood of a film can change either progressively over time or abruptly from scene to scene. The composer must have sufficient control over the music to allow both subtle and agile emotional change.

These are issues that we have had to consider while creating Metascore. The design of Metascore's user interface accommodates both structural synchronisation and composition control through the implementation of a number of specialised features. In contrast to the majority of film scoring applications currently available (Sony Creative Software Inc. 2008, Smartsound Software Inc. 2008), Metascore uses compositional algorithms to generate its musical material. This allows the user to set the parameters of the music, such as key structural points and compositional elements, in order to control the timing and mood of the score over time. Metascore then takes care of calculating how to fit the music to the score, a task usually undertaken by the composer.

The interface has been developed to facilitate our research of the underlying generative algorithms employed. This means that it is not intended to be end-user ready, which is apparent, for instance, in the fact that some parameters require Lisp expressions. Although this would be unsuitable as a final release, it allows us as developers to powerfully express the input for these algorithms. It helps us to discover what types of user input and functionality are effective and allows us to explore various possibilities for end-user interface design. The Metascore interface also allows us to rapidly test our ongoing work on the underlying compositional algorithms.

Similar Projects

There are a number of other projects which are related to our work on Metascore. Some of these include:

Both Cinescore (Sony Creative Software Inc. 2008) and Sonicfire Pro (Smartsound Software Inc. 2008) are commercial scoring tools which take the approach of adapting pre-recorded fragments or themes to video, achieved primarily by stretching or compressing passages to fit the length of a scene or section. The user outlines the structure of the scenes in the film with a series of markers and selects a recorded fragment which suits the mood they desire. While the features and implementation vary between the two programs, both are constrained to the library of available recordings and limit the user's creative input, yet provide a quick and easy way to create accompaniment for a film.

Scorebot (Pierce 2000) concerns the theory and design of an automated film scoring tool and, although it was not implemented, identifies many issues surrounding generative film scoring. Its design explores the manipulation of pre-composed music to be used as a film score by processing a given theme to align with a set of scene information and cues, such as the timing, mood and events associated with the scene. It is proposed that this be achieved through altering the compositional aspects of the theme, including the dynamics, modality, rhythms, tempo and dissonance of the themes.

Qsketcher (Abrams, Bellofatto, Fuhrer, Oppenheim, Wright, Boulanger, Leonard, Mash, Rendish, Smith 2001) is a composition environment that considers the creative workflow of composition (in the context of film music) and ways in which to enhance the role of the computer in this process. Its design addresses the representation of musical ideas, their context and how they relate to one another, in such a way as to support the cognitive association with composition and creative workflow.

Hyperscore (Farbood, Kaufman, Jennings 2007) is a graphical composition environment in which the user composes music by drawing various lines, curves, shapes and textures. The graphical information is then interpreted by Hyperscore's compositional algorithms (in accordance with musical motifs) to produce the composition. This gives non-musically inclined users the ability to express compositional intent in an abstract manner and to explore musical creativity. While not directly applicable to film composition Hyperscore is another significant commercial application in the generative music space and is one of the only tools which also investigates the generation of orchestral scores.

These applications all identify pertinent issues concerning the creation of music synchronised to film and the user interface design required to facilitate this. Each deals with these issues using various approaches, providing different advantages through their implementation.

Metascore overview

Metascore is being developed in the Impromptu environment (Sorensen 2005) and makes use of the Vienna Symphonic Library Special Edition for sample playback. It facilitates the experimentation of algorithmic processes for genera-

tive film music and allows the user to control parameters of the underlying generative algorithms that focus on high level artistic issues such as intensity, complexity, tension and the like. These higher level considerations are often referred to as meta structure (Xenakis 1992) and control of them as meta-level composition.

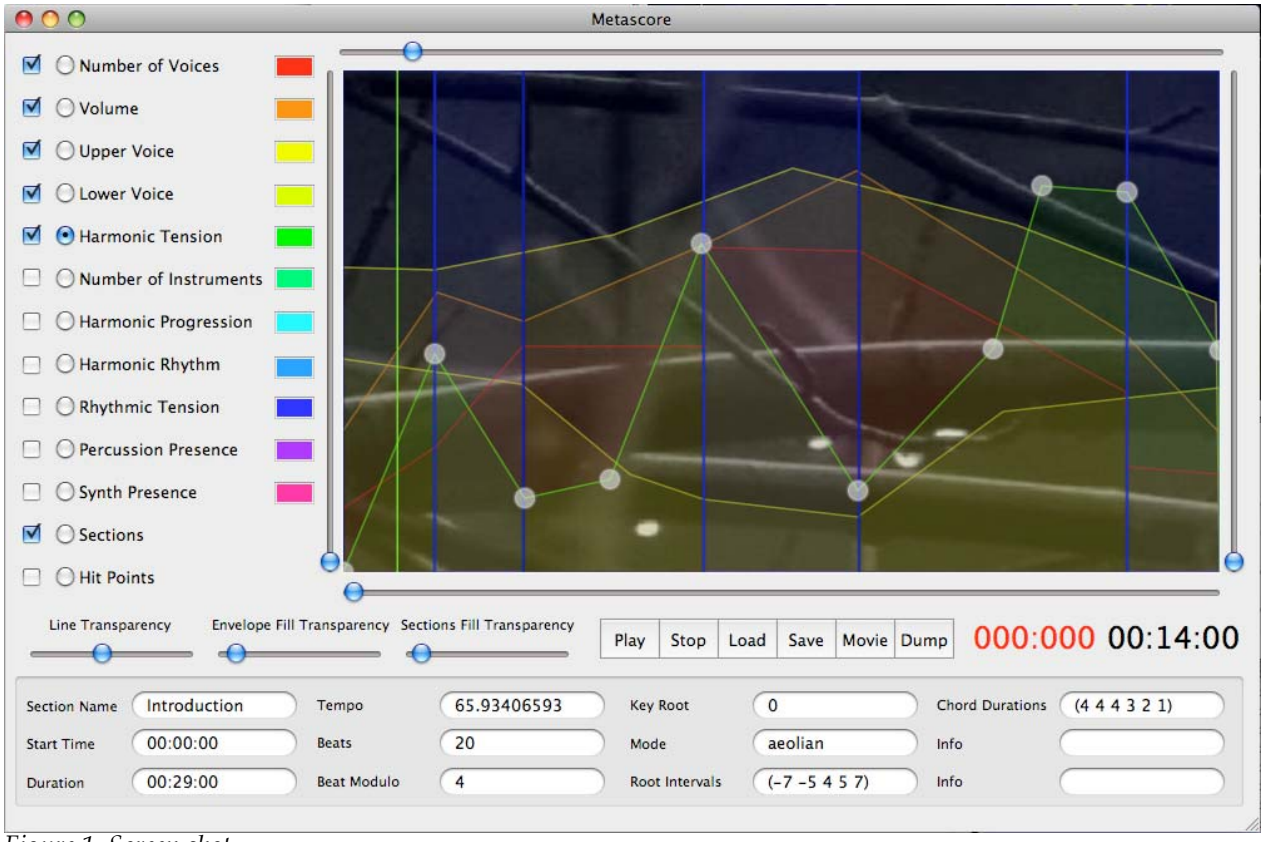


Figure 1. Screen shot.

Sections

Metascore enables the division of the score into sections which group together global parameters for specific parts of the film. Each section has its own set of values that are used to provide local state for the duration of the section. When combined, these sections outline the development of the entire composition over time. The main function of sections is to provide high-level musical structure with a cohesive relationship to the underlying film. Section markers are used to segment the film into cues. Metascore calculates bars, beats and tempi to ensure that natural musical boundaries always match cue points. Various parameters, including tempo, key and chordal information, are sectional and their settings remain constant for the duration of a given section.

In addition, Metascore provides a layer of parameter envelopes that allow for continual variation in parameters over time. Parameter values are interpolated between break points in the envelopes, which are used as input into the compositional algorithms that generate the music.

In figure 1, section markers are indicated by vertical lines on the video display area, which acts as a time line. The various parameter envelopes are shown overlaid on top of these vertical section markers. The position within, and resolution of, the time line is controlled by the sliders beside and above the display. The left side of the interface shows the musical parameters than can be controlled by envelopes and allows for them to be toggled on and off. The lower part of the display includes transport controls and fields for parameter values.

Tempo Snapping

The use of sections relates to events in the film on a macro scale. One important structural aspect of music, as distinct from film, is the grouping of regular repetitions of pulses into bars and beats. Film duration is usually measured in minutes, seconds and frames. In order to make the translation from the film world, based in irregular durations in frames, into the music world of more regular temporal patterns in beats, it is important to determine a musical tempo that will accommodate the requirements of both domains. Metascore allows the user to specify a tempo for

each section, but in such a way as to ensure that it contains a whole number of beats or bars. The specified tempo is adjusted to the nearest value which would accommodate a whole number of beats over the duration of the section. This calculation will usually push the tempo up or down by up to five beats per minute. If the new tempo is to far from the composer's intent, a change to the number of bars and beats can be made and the process repeated. When applied to all sections, this effectively synchronises the beats of the music with the structure of the film. These Tempo Snapping calculations occur whenever a section is created or modified.

Beat Modulo

A feature related to tempo snapping is Beat Modulo. This allows for the grouping of beats, so that a whole number of these groups fit within the duration of the section. Again, this is achieved by further modification of the tempo. This feature is useful for specifying the bar length used in the section (related to specifying a time signature), which prevents the use of an irregular number of beats and allows for further constraints over the musical length of a section while maintaining tight links to the structure of the film.

Sectional Harmonic Parameters

The remaining variables for a section are the key, mode, chord intervals (steps between sequential chords) and chord durations. These are important parameters that greatly influence the musical character of each section. For instance, the pitch scale used in a section can be easily defined by specifying the root note of the key and name of the desired mode.

Another useful feature is the ability to supply a list of possible root progression intervals and chord durations which are used as input to the underlying compositional algorithm. This allows the output of each section to be specified in more detail. These features can be used to support the linear structure of a film composition. When these parameters are used effectively, and in coordination with the other sections, it is possible to produce a well controlled harmonic basis for the overall composition.

Hit Points

Even with the use of sections to align the composition and film structures, there still may be cases where particular off-beat actions within a section of the film require musical accentuation. This is a situation commonly encountered in film composition and is known as a hit point. Metascore deals with hit points by subdividing a section at the point where this accentuation is needed and forcing the first beat of the nearest bar to lie on the hit point. Subdivisions created before and after the hit point are tempo snapped. Both subdivisions exist as a part of the same section and still share all of the other section characteristics.

We also plan to provide independent hit points, which are unrelated to any beat association or synchronisation, although at present our interest is in seeing how far we can push these tempo relationships.

Envelope Control

Metascore provides control over many compositional elements via continuously varying parameters. These are most frequently used to express compositional intent through features such as volume, orchestration, melodic continuity, texture, pitch compass and the like. The details and workings behind these elements and algorithms are discussed at greater length in related papers about generative music algorithms that Metascore can control (Sorensen & Brown 2008).

The control of these elements is given to the user as a series of break-point envelopes that consist of a custom number of editable nodes as shown in figure 2. These envelopes represent the value or state of a parameter over the duration of the entire film. Their values are used as input into the compositional algorithms which, in turn, generate the score on a note-by-note level in real-time. The various combinations and arrangement of these envelopes, in conjunction with the various section parameters discussed earlier, are the building blocks of any given composition.



Figure 2. Break point envelope in detail.

The envelopes themselves can be easily modified, allowing the user to create new nodes, drag around existing ones, or delete unwanted nodes from the break-point envelope. A key editing feature is the fact that the envelopes are overlaid on to the video. This makes for easy editing, as the user can drag the time slider to specific visual events and align envelope nodes with precision, ensuring that key compositional points match up with corresponding events in the film. The user can zoom in on an envelope for even finer control. Additional editing capabilities include the option to view multiple envelopes simultaneously, as well as to control the transpar-

ency of the overlayed envelopes. Envelope display options can be used to enhance the editing process by putting related envelopes into simultaneous view and editing them in concert. Envelopes can also be used to depict a meta "score". By increasing the transparency of the envelopes it is possible to view, and hear, the interaction between the compositional parameters in real-time.

With effective arrangement of the various envelopes over the duration of the film, the user has essentially composed the film's score using the contours of the envelopes. This method of composition emphasises a balance between control and efficiency. Control of the composition on a note-by-note level is delegated to the generative algorithms and the user engages with envelope control of compositional elements over time.

The intention behind these compositional envelopes is that they can influence the emotional content of the generated music. For instance, the envelopes could be arranged to indicate action by using a high rhythmic tension value, high harmonic tension for suspense, and a large number of instruments for a more general crescendo. But these examples are overly simplistic. A more accurate description of their use would include the careful arrangement of all the parameters working together to create a specific mood. Potentially, the more parameters that were available, the greater the possibilities for more complex emotions. The integration of parameters to achieve specific moods is an area we are still investigating through the development of Metascore. However, even with the current editing capabilities, it is quite possible to arrange the compositional elements to accurately coincide with the film's emotional content and to create some basic moods in the music to match. The governing idea is that with the interface features described above, the user will be able to sculpt the mood of the music around the structure of the film, using the compositional envelopes in any combinations they see fit.

Using Metascore

The method of composition provided by Metascore is reliant on both the ability to compose accurately with the interface and on the quality of the interpretation by the algorithms. The accurate realisation of user compositional intent has been one of our main concerns in the development of Metascore's interface. This has lead us to provide the user with meta-level control over the composition which allows us, as researchers, to explore the full range of output that Metascore is capable of generating.

The inherent limitation of this method is the fact that it takes away the absolute control of composition usually available to the composer. However, this limitation is counter-balanced by the efficiency of synchronisation between music and film, the ability to compose on a structural level, the ability for music to follow edits in the film, and the ability to rapidly and interactively experiment with compositional options.

Conclusion

This paper has introduced a new interface for film composers designed to support the structural and emotional support for the actions and events which occur visually in the film. In order to fulfil these roles successfully, synchronisation between the structure of the music and the structure of the film is vital. As film structures are irregular in comparison to common musical structures, the musical structures for film require special consideration. The design of the Metascore interface addresses these issues for the generative music composer and provides a platform for efficient experimentation with a variety of parametric settings and thus a quick exploration of the compositional space occupied by any particular compositional algorithm.

We have discussed how the use of sections can support the kind of linear development which is effective in the composition of film music, particularly for harmonic and metric structure. We have described how Metascore makes use of section breaks and parameters envelopes to align musical features with the structure of a film and to potentially craft the emotional content of the music in relation to the visuals.

The future development of Metascore lies in other areas inherent to film composition, such as the use of pre-composed themes and thematic development. As this development continues, the interface design of Metascore will evolve also to allow for the effective experimentation of generative music for film.

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